

Module 1 — How Orbits Work (Orbit Academy v5.5)

Printable edition. The interactive version — live simulator, self-checking questions — is the course website; open the matching tut/worksheet pages there. Dynamic diagrams appear here as bracketed figure notes.

What you will learn

- why a satellite doesn't fall down — it's **falling sideways** fast enough to keep missing the Earth.
- how your **release speed** decides the whole shape of the path — a crash, a clean circle, a stretched **ellipse**, or an escape.
- why an object races through **perigee** (closest) and crawls at **apogee** (farthest), and why **farther out means slower**.
- what **geosynchronous** and **geostationary** really mean, and why a 24-hour orbit can hang over one spot on Earth.
- how **inclination** tilts an orbit — and why launch sites like Florida and Vandenberg point in the directions they do.
- a gut sense of **scale**: just how close “space” (and the ISS) actually is.

Quick-start tutorial

An **orbit** sounds exotic, but it's really just falling. Gravity is always pulling a satellite straight down toward Earth. What keeps it up isn't some anti-gravity trick — it's **side-ways speed**.

Picture this: Fire a cannonball off a tall mountain. Lob it gently and it arcs down and hits the ground. Fire it harder and it lands farther away. Fire it hard enough and the ground *curves away beneath it just as fast as it falls* — so it keeps missing the Earth, forever. That's an orbit: falling sideways so fast you never come down.

Speed sets the shape

At any given height there's exactly **one speed** that gives a perfect **circle**. Go a little slower and you dip toward Earth on the far side (too slow, and you crash). Go faster and you swing out wide, tracing a stretched-out oval called an **ellipse**. Faster still — about 10.7 km/s from the 600 km release height you'll use here — and you escape Earth entirely.

[Figure: Same launch point, faster each time: too slow and the far side dips into Earth — it re-enters (dashed red); the right speed gives a circle (green); more speed opens a bigger ellipse (blue) that swings far out on the opposite side.]

On an ellipse the object moves **fastest when it's closest (perigee)** and **slowest when it's farthest (apogee)**. And the higher an orbit is, the slower it goes overall — a low satellite whips around in about 90 minutes, while a far-out one at **GEO** takes a full 24 hours, keeping pace with Earth's spin so it hovers over one spot.

A couple more words you'll meet: **prograde** means going with Earth's spin (the easy, default direction); **inclination** is how much the orbit is tilted — measured against the

plane of Earth's equator — which decides how far north and south the satellite travels.

Getting around the simulator

Click ↗ **Simulator** (top-right of this page) to open the 3-D view in its own window, and set it beside this worksheet. You steer the *camera* with your mouse and keyboard — you're moving your viewpoint, not the satellites. The same controls work in every module, and the **Cheat sheet** button keeps them handy.

[Figure: The camera controls, identical in every simulator: left-drag to pan, Shift-drag (or right-drag) to rotate, **pinch the trackpad or roll the mouse wheel to zoom**, arrow keys to rotate, “,” / “.” to slow/speed time, R to reset the view.]

Try each one now: drag to slide the view, Shift-drag to spin around Earth, and roll the wheel to zoom. Then come back and start the exercises below. If you ever lose your bearings, press **R**.

PART A • Drive the simulator

Four tools to learn first: the MOUSE (drag to pan, wheel to zoom, Shift-drag to rotate), the ARROW KEYS (steady rotation), the , and . keys (time flow), and the control board on the left (switches, sliders, INJECT). Nothing here can be broken — press R to reset the view.

Exercise 1 — Move the view: pan, zoom, rotate

Before you can learn anything from the simulator, you have to be able to look around inside it. The single most important thing to get straight: the 3-D scene is a fixed little universe, and you are a **camera** flying around it. When you drag or zoom, the Earth and satellites never move — *you* move. Beginners often feel like they're shoving the planet around; you're not. You're just changing where you float and which way you're facing.

There are only three moves, and they're the same in every module. **Pan** slides your viewpoint sideways. **Zoom** flies you closer or farther. **Rotate** swings you around to see the scene from a new angle — over a pole, from the side, wherever you like. Getting fluent with these three now means that later, when the physics gets interesting, you can always find the best angle to watch it from.

Nothing you press can break anything. If you get twisted around and lose the Earth, the **R** key instantly snaps you back to the starting view — so explore freely.

Try it in the simulator:

Open the simulator — a  **controls card** greets you on the first run (the  button in the **View** panel brings it back any time). Now do each move deliberately, with a target: **left-drag** to slide the view so Earth sits off to one side; **roll the wheel or pinch** until the whole ISS orbit just fits, then zoom right in until the planet fills the screen; **Shift-drag** (or right-drag) to swing underneath and look up at Earth from below. Then take your hands off the mouse and use the **arrow keys**: hold ◀ or ▶ to spin around the equator, and ↑/↓ to tip over the pole until the ISS orbit is exactly **edge-on** — a straight line across the planet. Finally press **R** and watch the view snap home.

What to look for: you are moving your viewpoint, not the Earth — the planet and satellites hold still while you fly around them. The arrow keys give a slow, steady rotation that is far better than the mouse for lining up an exactly edge-on view, which is how you will judge the *tilt* of an orbit later in this module.

While you watch, think about:

- When you drag, does the Earth actually move, or does it just fill a different part of your screen?
- From how far out can you still see the whole scene at once? How close can you get to the surface?
- Which single move — pan, zoom, or rotate — do you reach for most to get a shot you like?

Check your understanding: In the simulator, what are you actually moving when you drag and zoom?

- The Earth, which you swing around while the satellites hold still in the window
- Your camera — the objects stay put while your viewpoint moves around them
- The orbits themselves, which tilt and stretch as you drag across the window
- Both the view and the orbits, since display and physics share one frame

Exercise 2 — Turn objects on and off

The simulator can show several real families of satellites at once — the ISS, Starlink, GPS, the GEO belt, and more. Each one is a **checkbox** on the left, and each checkbox is an independent on/off switch for a whole population of real objects, drawn to true scale.

The trick to actually learning from these is discipline: turn them on **one at a time**. With everything showing at once the screen becomes a confusing tangle of rings. Isolating a single layer lets you study its height, its speed, and its shape without distraction — then you hide it and bring up the next. Toggling a layer off never deletes anything; it just tucks it out of view until you want it back.

Try it in the simulator:

On the left, under **Real orbits (for scale)**, tick **GPS** and then **GEO** on, then turn them back off. Each checkbox adds or removes a whole population of real satellites.

What to look for: the checkboxes are independent layers you can combine freely — but a single layer, shown alone, is far easier to study.

While you watch, think about:

- With both GPS and GEO on, can you tell at a glance which ring is which?
- Does turning a layer off destroy those satellites, or just hide them?
- When you want to study one orbit closely, is it easier with everything on, or just that one layer?

Check your understanding: What does each checkbox under “Real orbits” do?

- Deletes that group from the run, so you must reload the page to bring it back
- Shows or hides that group of real satellites; each group toggles on its own
- Swaps the camera to a preset angle chosen to frame that particular group
- Injects a fresh copy of that group using the altitude set by the sliders

Exercise 3 — Speed up and slow down time

Real orbits are slow. The fastest satellites here take about 90 minutes to circle the Earth; the far-out ones take a full day. If the simulator ran at real speed you'd stare at a frozen picture — nothing would visibly move. So the sim lets you **warp time**, compressing hours into seconds so you can actually watch motion happen.

Speeding time up doesn't change the physics one bit — the orbits are exactly as accurate fast as they are slow. It only changes how quickly you watch them play out. A readout at the top-left tells you the current rate (for example "minutes per second" or "hours per second"), and you can always tap **,** a few times to slow way down — to 1/32 speed — and study a moment closely.

Try it in the simulator:

Press **.** (period) a few times to speed up time and watch satellites move; press **,** (comma) to slow it back down — the time-rate indicator is top-left. Now use it with a purpose: speed time up until the **ISS** takes roughly **ten seconds** to go once around, and note the rate you needed. Then tick **GEO** on and try to watch one of those satellites complete a lap: you will have to speed time up a lot further. Slow it back to $\times 1$ when you are done and notice how nearly frozen everything looks.

What to look for: the satellites start to visibly sweep along their paths as you warp time, and the top-left readout shows how much real time passes per second. A LEO satellite needs only a modest speed-up to be watchable; GEO needs far more, because one GEO lap takes a whole day against about 90 minutes in LEO. Time warp is not a gimmick — without it, most of what this course teaches happens too slowly to see.

While you watch, think about:

- At real speed, could you even tell the satellites were moving?
- Does speeding up time change the shape or size of an orbit, or only how fast you watch it?
- How much time warp do you need before a slow, far-out orbit visibly moves?

Check your understanding: Why will you almost always run the simulator with time sped up?

- The integrator is only accurate at high warp, where it takes steadier steps
- Real orbits take 90 minutes to 24 hours — far too slow to watch in real time
- Speeding the clock cuts the work per orbit, which keeps the frame rate smooth
- Orbits only close into repeating paths once the clock runs faster than real time

Exercise 4 — The control board: switches, sliders, and INJECT

Everything you *do* in this simulator happens on the panel down the left-hand side, and it comes in three kinds. The **tick boxes** at the top are switches: each one turns a whole real satellite population on or off. Below them, three **sliders** describe an object you are about to launch — how high you release it, how fast you throw it sideways, and how tilted its path will be. And the **◇ INJECT OBJECT** button actually launches it.

The order matters, and it catches people out: the sliders only decide what the *next* object will be. Moving a slider does nothing to something already in orbit — an object in flight

is on its own from the moment you release it, exactly like the real thing. To try a different speed, you set the slider and inject again.

Predict first: If you inject an object and then drag the speed slider, do you expect the object already up there to change its path?

Try it in the simulator:

Set **release altitude** to 600 km, leave the speed where it is, and press $\diamond$ **INJECT OBJECT**. Now drag the **tangential speed** slider well to the left and watch the object you already launched — does it react? Press $\diamond$ **INJECT OBJECT** again to launch a second one at the new speed, and compare the two paths. Then use **clear my objects** to wipe the slate.

What to look for: the sliders only load the *next* shot — the object already in flight ignores them completely and keeps coasting on the speed it was given. Injecting again adds a second, different orbit alongside the first, which is exactly how you will compare speeds in Part C.

While you watch, think about:

- Why is it physically right that a slider cannot change an object that has already been released?
- What would you have to do, in real life, to change an orbit that is already flying?

Check your understanding: You inject an object, then move the speed slider. What happens to the object already in orbit?

- Its orbit stretches or shrinks at once to match whatever the slider now reads
- Nothing — the sliders arm the NEXT injection; one in flight keeps its speed
- It keeps its shape but circles faster, since the slider sets orbital speed
- It vanishes, because changing the slider invalidates its released state

PART B • Meet the real orbits, one at a time

Each checkbox in the sim is a real population of satellites, shown to scale. Turn them on ONE at a time and study each before moving on.

Exercise 5 — How high is space, really?

Ask most people to picture the space station and they imagine it way out among the stars, halfway to the Moon. The reality is startlingly different. The **International Space Station** orbits at about 420 km — and Earth’s radius is roughly 6,400 km. That means the ISS flies at only about **7% of an Earth-radius** above the ground. On a globe drawn to scale, it practically skims the paint.

This scale surprise matters. “Space” isn’t some far-off frontier; the edge of it is closer than a lot of road trips. Seeing the ISS ring hug the globe rewires the intuition you’ll need for everything else in this module — because *how high* an orbit is turns out to control almost everything about it.

Predict first: Before you look: picture the space station in your mind. On a to-scale globe, where do you expect its ring to sit — way out toward the Moon, halfway up, or hugging the paint of the surface?

Try it in the simulator:

Turn OFF every checkbox except the **Space Station (ISS)**. Look at the globe. Then turn the ISS off and flick on **GPS** for a second to see how much higher it sits, and switch back — the contrast makes the ISS's low altitude obvious.

What to look for: the ISS ring sits right against the surface — its altitude is only about 7% of Earth's radius.

While you watch, think about:

- Compared to the size of the Earth, does the ISS look “out in space” or right against the surface?
- If Earth were a basketball, about how far off its surface would the ISS ring float?
- Does this match the mental picture you had of where the space station lives?

Check your understanding: On the to-scale globe, where does the ISS orbit?

- Far out, about a third of the way to the Moon
- Skimming just above the surface
- At the same height as GPS
- Beyond the GEO belt

Exercise 6 — How far away is the ISS overhead?

Here's a way to feel that 420 km in everyday terms. When the ISS passes directly over your head, it's about **260 miles** straight up — roughly the distance from Washington, DC to New York City. If you can drive somewhere in an afternoon, that's about how far “space” is when it's right overhead.

People routinely overestimate this by a factor of ten or more, guessing thousands of miles or even reaching for the Moon. Anchoring the number to a familiar trip makes the truth stick: the busiest orbits in the world are shockingly close overhead.

Predict first: Before you check the answer: if the ISS were passing straight over your head right now, how far away would you guess it is — a few miles, a couple hundred miles, or thousands?

Try it in the simulator:

Keep the ISS shown. Picture it passing directly over your head at ~420 km (~260 miles), and match that to a trip you know before revealing the answer below.

What to look for: straight up, the ISS is only about a DC-to-NYC distance away — closer overhead than most cross-country drives.

While you watch, think about:

- Before reading the answer, what would you have guessed the overhead distance was?
- Which everyday trip is closest to 260 miles for you?
- Why do you think people picture the ISS as being so much farther away than it is?

Check your understanding: When the ISS is directly overhead, roughly how far from you is it?

- DC to Arlington (~5 miles)

- DC to New York City (~225 miles)
- DC to Los Angeles (~2,300 miles)
- To the Moon (~240,000 miles)

Exercise 7 — LEO: one low, fast orbit

Low Earth Orbit (LEO) is the crowded neighborhood just above the atmosphere — a few hundred kilometers up, where the ISS and most imaging satellites live. It's low, and here's the key pattern you'll see again and again in this module: **low means fast**. Down close to Earth, gravity is strong, so a satellite has to move very quickly to keep "falling around" the planet instead of into it.

A satellite at 550 km whips around the whole Earth in only about **90 minutes**. That's roughly 16 laps every single day. Watch it against the globe and you'll see it both hug the surface like the ISS and race along noticeably faster than anything higher up.

Predict first: Before you speed up time: this orbit is very low, close to Earth where gravity is strong. Do you expect it to move fast or slow — and would one lap take more like 90 minutes or a whole day?

Try it in the simulator:

Turn off the ISS and turn on **LEO demo** (one equatorial orbit at 550 km). Speed time up with . and watch it go around. Later, when you turn on the far-out GEO ring, come back and compare how much faster this low one races.

What to look for: it hugs the Earth like the ISS and, at this low altitude, whips around fast — one lap in about 90 minutes.

While you watch, think about:

- How close to the surface is it compared to Earth's overall size?
- If one lap takes about 90 minutes, roughly how many laps would it complete in a day?
- Does it look faster or slower than you expected for something orbiting the Earth?

Check your understanding: A satellite in low Earth orbit (~550 km) completes one orbit in roughly...

- About 24 hours, the same as the Earth turning once beneath it
- About 90 minutes, so it laps the planet roughly 16 times a day
- About 1 minute, since low orbits are the fastest ones there are
- About a week, because it has so much further to travel than GEO

Exercise 8 — Starlink: a whole shell of inclined orbits

A single orbit is just one ring around the Earth — and one ring only ever passes over the same narrow band of ground. That's fine for some jobs, but if you want to beam internet to *everyone*, everywhere, all the time, one ring won't do. You need coverage over the whole planet, always.

Starlink solves this with a **shell**: hundreds (eventually thousands) of satellites spread across many orbits **tilted at different angles**. That tilt is called **inclination** — orbits at 53°, 70°, and 98° reach far north and south instead of staying over the equator. Layered

together, they blanket the globe so that from anywhere on Earth there's always a satellite overhead to connect to. When you look down over a pole you'll see the criss-crossing weave; from the side you'll see how the tilts stack into a thick shell.

Predict first: Before you turn it on: to give fast internet to everyone on Earth all the time, do you think one big ring would work — or would you need many orbits, and if so, why not just stack them all over the equator?

Try it in the simulator:

Turn off LEO and turn on **Starlink** (~400 sats in inclined shells at 53°/70°/98°). Rotate the view to look straight down over a pole, then swing around to a side-on view and compare — the pole view shows the criss-cross weave, the side view shows the tilts stacking into a shell.

What to look for: instead of one ring it's a dense shell of many tilted orbits, blanketing the globe so someone is always overhead.

While you watch, think about:

- Why would one equatorial ring, like the LEO demo, leave much of the planet with no coverage?
- What does tilting the orbits at different angles let the shell reach that a flat equatorial ring can't?
- Why might a company need hundreds of satellites instead of just a handful?

Check your understanding: Why does Starlink use hundreds of satellites in tilted (inclined) orbits rather than a few equatorial ones?

- Tilted orbits are cheaper to reach, so more satellites fit one launch budget
- A tilted shell covers the whole globe, poles included, one always overhead
- Equatorial orbits are already full of GEO traffic, so the belt has no room
- Inclined orbits decay more slowly, so the constellation lasts much longer

Exercise 9 — GPS: mid-altitude, half-day orbits

The GPS constellation lives in **Medium Earth Orbit** — about 20,200 km up, roughly 48 times higher than the ISS. That's no accident. To fix your position on the ground, a receiver needs to see several satellites at once; and the higher a satellite sits, the larger the patch of Earth that can see it at any moment. From way up there, each GPS satellite covers a huge footprint, so a modest fleet keeps several in view from anywhere on the planet.

And remember the pattern: higher means slower. Where a LEO satellite laps the Earth in 90 minutes, a GPS satellite takes about **12 hours** per orbit — half a day. When you turn it on you'll have to zoom way out to find it, and you'll notice it drifting along far more lazily than the low, fast orbits you watched earlier.

Predict first: Before you zoom out: GPS orbits about 48 times higher than the ISS. Based on the "low means fast" pattern you just saw, do you expect it to move faster or slower than the LEO ring — and why put navigation satellites way up there at all?

Try it in the simulator:

Turn off Starlink and turn on **GPS (MEO)** — 20,200 km, 6 planes at 55°. Zoom out to find it. For a direct comparison, flick the **LEO demo** back on alongside it for a moment and watch the two paces side by side, then turn LEO off again.

What to look for: GPS sits far higher than LEO and moves noticeably slower, taking about 12 hours per orbit.

While you watch, think about:

- How far do you have to zoom out for GPS to fit on screen, compared to the ISS?
- From that height, how big a slice of the Earth do you think one satellite can “see”?
- Why might navigation be the one job where flying high is worth the extra effort to get there?

Check your understanding: GPS satellites orbit much higher than LEO (20,200 km). One benefit of that altitude is...

- Higher satellites move faster, so each sweeps past a receiver more often
- From higher up each satellite sees a far larger patch of Earth at once
- The thinner air up there removes drag, so no station-keeping is needed
- Higher orbits sit above the radiation belts, keeping the electronics safe

Exercise 10 — GEO: the far ring that keeps pace with Earth

Push all the way out to about 35,786 km and you reach the **geostationary belt** (GEO) — the far ring, and the slowest orbit in this whole picture. At that exact height a satellite takes one **sidereal day — 23 h 56 min, exactly one Earth rotation** — to complete one lap.

That’s the magic number. Because the satellite goes around once per day while the Earth spins once per day, the two stay in lockstep — the satellite effectively **hovers over one spot** on the ground. That’s why TV dishes and weather satellites live here: point your antenna once and it never has to move. For now, just take in how high and how slow GEO is compared to the racing LEO ring; you’ll unpack the “hovering” trick fully in Part F.

Predict first: Before you speed up time: GEO is the highest orbit of all here. How slow do you think it will be — and can you guess why its exact period of one day turns out to be so useful?

Try it in the simulator:

Turn off GPS and turn on **GEO** (35,786 km, equatorial). Zoom out — it’s the far ring. Speed time up, then briefly turn the **LEO demo** back on and watch both together: count roughly how many times the low ring laps while GEO barely creeps along.

What to look for: GEO is the highest and slowest of these, taking a full 24 hours per orbit — exactly one Earth day.

While you watch, think about:

- What’s special about an orbit whose period equals exactly one day?
- Side by side, how much slower does GEO crawl compared to the LEO ring you watched?
- If the satellite circles once a day and Earth spins once a day, what would the satellite look like from the ground?

Check your understanding: What is distinctive about the GEO ring's orbital period?

- It is the shortest period on the display, which is why the ring looks static
- It equals one Earth day, so it keeps pace with the ground turning below
- It drifts slowly over weeks as the equatorial bulge tugs the satellites along
- It is exactly 90 minutes, matching the low-orbit satellites further down

PART C · What makes an orbit? (inject your own at 600 km)

Now build orbits yourself and watch how release speed sets the whole shape.

Exercise 11 — Too slow: smash into Earth

Now you get to build orbits yourself. The core idea of this whole module is that at any given height, the **speed you release an object** at decides the entire shape of its path. Get a feel for that by starting with the failure case: going too slow.

An orbit only works if you're moving sideways fast enough that the ground curves away beneath you as fast as you fall. Release too slowly and you can't "miss" the Earth — you simply fall back into it. The path bends into a steep, stretched ellipse whose far side dips *below the surface*, so the object **re-enters the atmosphere** and comes down. This is exactly what a suborbital lob looks like: up, over, and back down into the ground.

Predict first: Before you inject: if you release well below the speed needed to orbit, what do you think the path will do — settle into a lower circle, or fall back and hit the Earth? And where would it strike, near your release point or the far side?

Try it in the simulator:

Set **release altitude = 600 km**. Set **speed ≈ 4.5 km/s** and click **INJECT OBJECT**. Then nudge the speed up to ~ 6 km/s and inject again — watch the far side lift a little but still dip below the surface, so it's still a crash.

What to look for: the orbit is a steep ellipse whose far side plunges below the surface — it re-enters the atmosphere and comes down.

While you watch, think about:

- Where does the path first touch the Earth — near your release point, or on the far side?
- What do you think would happen if you nudged the speed up a little from 4.5 km/s?
- Why can't an object this slow ever "miss" the ground and keep going around?

Check your understanding: At 600 km, injecting at only 4.5 km/s (well below circular) gives you...

- A slightly lower circular orbit, since less speed simply means a smaller circle
- An ellipse whose far side dips below the surface — it re-enters the atmosphere
- An escape trajectory, because releasing off-speed always opens the orbit up
- A hover at 600 km, with gravity and the release speed balancing each other

Exercise 12 — Just right: a circular orbit

At any altitude there is exactly **one special speed** — the **circular speed** — that threads the needle between falling back and flying out. Hit it exactly and the object falls toward Earth at precisely the rate the horizon curves away, so it neither climbs nor dips. The result is a perfect circle at a constant height.

This is the balance point everything else is measured against. A hair slower and the far side sags into an ellipse that dips lower; a hair faster and it bulges outward into an ellipse that climbs. There's nothing magic about the circle except that it's the one speed where those two tendencies cancel out perfectly.

Predict first: Before you inject: you saw 4.5 km/s crash. Do you think there's a single exact speed that gives a perfect circle, or would a whole range of speeds all work equally well?

Try it in the simulator:

Raise the speed to the “ $\approx$ **circular**” value (~ 7.6 km/s at 600 km) and inject. Then deliberately miss it — inject at 7.2, then 8.0 km/s — and compare: notice how even a small miss turns the clean circle into a visible oval.

What to look for: you get a clean circle — the object falls toward Earth exactly as fast as the horizon curves away.

While you watch, think about:

- How precisely do you have to hit the circular speed before the ring stops looking like an oval?
- Compared to the crash case, how much more speed did it take to hold a full orbit?
- Is a circle really “special,” or just the one speed where climbing and dipping cancel out?

Check your understanding: What makes an orbit circular rather than elliptical?

- Releasing at exactly the local circular speed for that altitude
- Releasing as slowly as you can without the orbit dipping into the atmosphere
- Releasing at escape speed, the one speed that gives a perfectly closed path
- Any release speed will do — the shape is set by the altitude, not the speed

Exercise 13 — Faster: an ellipse that climbs

Now go the other direction from the crash — release **faster** than circular. The extra speed carries the object outward, so instead of falling back it swings up and away on the far side of the Earth, tracing a stretched oval, an **ellipse**.

Here's the important twist: the point where you let go becomes the **lowest** point of the new orbit — the **perigee**. The object climbs away from there to a high point on the opposite side, called the **apogee**, then falls back down to your release point and does it all again. More release speed means a higher, farther apogee. (Compare this to the too-slow case, where the release point became the *high* point and the far side dipped down instead.)

Predict first: Before you inject: when you release FASTER than circular, which side of the orbit do you think swings outward — and will your release point end up as the high

point or the low point of the new orbit?

Try it in the simulator:

Now push the speed **above circular** (e.g. 8.5 km/s) and inject. Then bump it higher — try 9.5 km/s and inject again — and compare how much farther the apogee balloons out with the extra speed.

What to look for: your release point becomes the perigee (low point) and the far side balloons out to a high apogee.

While you watch, think about:

- Is your release point the highest or the lowest point of the new orbit?
- As you add more speed, how does the far side (the apogee) change?
- How is this different from the too-slow crash, where the far side dipped down instead of up?

Check your understanding: You release FASTER than circular. Your release point is now the orbit's...

- Apogee (farthest point)
- Perigee (closest point)
- It stays circular
- Escape point

Exercise 14 — Speed changes around the orbit

On a circle, a satellite moves at a steady pace the whole way around. On a stretched ellipse it does something you can see with your own eyes: it **speeds up and slows down** over the course of each lap. It moves **fastest at perigee**, its closest point to Earth, and **slowest at apogee**, its farthest.

The reason is gravity. As the object falls inward toward perigee it picks up speed, like a ball rolling downhill; climbing back out to apogee it coasts and slows, like the same ball rolling up. This is one of Kepler's great discoveries — an orbiting object sweeps out equal areas in equal times — but you don't need the math. On a good elongated orbit the change is obvious: a fast dash past the planet, then a long, lazy crawl way out at the top.

Predict first: Before you watch the marker: on a stretched ellipse, where do you think the object moves fastest — swooping close to Earth, or way out at its farthest point? And where will it seem to linger longest?

Try it in the simulator:

Clear your objects, then inject a good demonstration ellipse: **altitude 1000 km, speed 9.75 km/s, inclination 20°**. Bump time warp with . and watch the marker. Then inject a nearly circular orbit for contrast and note how *that* marker keeps a steady, unchanging pace all the way around.

What to look for: it visibly races through perigee (close to Earth) and crawls at apogee (far away) — the speed difference is dramatic on this elongated orbit.

While you watch, think about:

- At which point of the orbit is the marker moving fastest — and where does it seem to stall?
- Does the object spend more of its time near Earth or way out at apogee?
- What is it about gravity that makes the object gain speed as it falls inward?

Check your understanding: Where on an elliptical orbit does the object move fastest?

- At apogee (farthest)
- At perigee (closest)
- Same speed everywhere
- At the midpoint

Exercise 15 — Inject a fan: one point, seven speeds

You’ve now seen the three cases one at a time — too slow crashes, just right circles, faster balloons out. The **fan** tool puts all of them on screen at once so the whole idea clicks in a single picture.

It releases seven objects from the **exact same point**, each at a slightly different speed. Because they share a starting point, every one of their orbits must pass back through that point once per lap — but their sizes differ wildly. The slow ones trace tight inner ellipses; the fast ones swing way out into big outer ellipses; and somewhere in the middle sits the near-circle. Watching the nested family bloom out of one shared point is the clearest way to feel that **release speed alone sets the shape**. One heads-up: at these settings the slowest one or two members of the fan are too slow to stay up at all — the sim draws each as a red “ RE-ENTERS” **arc that ends at Earth’s surface** (not a full ellipse; the rest of the ellipse would be underground, and you can’t fly through a planet). That’s expected, not a bug — it’s the “too slow → crashes” case showing up right where it should.

Predict first: Before you fire the fan: seven objects will leave the exact same point at different speeds. What do you think their orbits will have in common — and will the fastest one trace the biggest orbit or the smallest?

Try it in the simulator:

Clear your objects. Pick an altitude (try 1000 km) and a middle speed, then click **inject a fan (7 speeds, centered on your speed)**. Bump time warp with . and watch all seven at once. Then clear and fire a second fan from a different altitude and compare — the nested pattern shows up the same way every time.

What to look for: all seven leave the same point but at different speeds, tracing a nested family of orbits that all pass back through that shared release point.

While you watch, think about:

- What do all seven orbits have in common no matter how big or small they get?
- Which released faster — the ones tracing the biggest orbits, or the smallest?
- Can you spot the one that’s closest to a perfect circle among the family?

Check your understanding: In the fan, seven objects leave the exact same point at different speeds. What do their orbits have in common?

- They are all identical circles, sharing one release point and altitude
- They all pass back through the release point but reach very different distances

- They never come near one another again after the moment of release
- The fastest traces the smallest orbit, because speed pulls it in tighter

PART D • Too much speed: escape

Push past escape velocity and watch a bound ellipse open into a one-way hyperbola.

Exercise 16 — Too much speed: escape (an unbound orbit)

So far, faster release has meant a bigger and bigger ellipse — but the object always came back. There's a limit to that. Push the speed high enough and the object gains more energy than Earth's gravity can ever claw back, and it leaves for good. That threshold is **escape speed**: about 11.2 km/s right at the surface, and a bit less — around 10.4 km/s — up at 1000 km, because gravity is a little weaker up there.

Below escape speed the path is **bound**: a closed loop (circle or ellipse) that repeats forever. Above escape speed the path **opens up** — it stops being a closed loop and becomes a **hyperbola**, a one-way curve that heads outward and never returns (exactly at escape speed it's a parabola, the knife-edge case). The object leaves its release point once and is simply gone.

The panel also reports an **eccentricity** — a single number for the orbit's shape: 0 = a circle, between 0 and 1 = an ellipse, and 1 or more = an open escape path (1 exactly is a parabola, above 1 a hyperbola). Watch it climb toward and past 1 as you cross escape speed.

Predict first: Before you inject: you've seen faster release make bigger and bigger ellipses that always return. Do you think there's a speed past which the object never comes back at all — and what would that path look like, still a closed loop or something open?

Try it in the simulator:

Clear your objects. Set **altitude 1000 km** and push the **speed** slider all the way up to **11 km/s**, then **INJECT**. (Escape speed at 1000 km is ~10.4 km/s, so 11 is above it.) Watch it head out and read the panel. Then drop to ~10 km/s and inject again — that one stays a closed loop, letting you find the knife-edge between coming back and leaving forever.

What to look for: the path no longer closes into an ellipse — it's an open hyperbola that leaves the release point and never comes back.

While you watch, think about:

- Does this path ever close back on itself the way an ellipse does?
- Drop the speed to about 10 km/s and re-inject — does it stay bound? How fine is the line between coming back and leaving forever?
- What does the panel report for the eccentricity above escape speed, versus for a normal orbit?

Check your understanding: You inject an object above escape speed. How is its path different from a normal orbit?

- It is a smaller, tighter circle, since the extra speed pulls it closer in
- It is an open curve — a hyperbola, eccentricity > 1 — and it never returns

- It is a perfect circle, because escape speed is the exact circular condition
- It crashes almost immediately, as too much speed drives it back into the Earth

PART E · Direction & the speed-distance law

Exercise 17 — Prograde vs. retrograde

An orbit doesn't just have a speed and a shape — it also has a **direction** of travel around the Earth. Seen from above the North Pole, the Earth spins **counter-clockwise**. An orbit that goes the same way, counter-clockwise with the spin, is called **prograde**. One that goes the other way, against the spin, is **retrograde**.

Prograde is the default for a simple reason: it's cheaper. A rocket launching eastward is already moving with the ground's rotation before it even lifts off, so it gets a free head start toward orbital speed. Going retrograde means first canceling all of that borrowed motion and then building speed the hard way — so it's reserved for missions that specifically need it.

Predict first: Before you read the note: if a rocket could launch either with Earth's spin or against it, which do you think saves fuel — and why would the ground's own motion give one direction a head start?

Try it in the simulator:

Read the "Prograde vs. retrograde" note in the sim. Earth spins counter-clockwise seen from the North Pole.

What to look for: a prograde orbit travels the same counter-clockwise direction the Earth itself is spinning, so it borrows that spin for free.

While you watch, think about:

- Which way is the Earth turning when you look straight down at the North Pole?
- Why would launching with the spin (eastward) save fuel compared to launching against it?
- Can you think of a reason a mission might deliberately choose the more expensive retrograde direction?

Check your understanding: An orbit traveling the SAME direction as Earth's spin (CCW from the North Pole) is called...

- Retrograde
- Prograde
- Polar
- Geostationary

Exercise 18 — Farther out = slower

One of the most important patterns in all of orbital mechanics is also one of the most counter-intuitive: **the farther out an orbit is, the slower it moves**. Far from Earth, gravity is weaker, so it takes less speed to stay in orbit — and with less speed, the trip around is slower still because the loop is also much bigger.

The numbers make it concrete. A low LEO satellite screams along at about 7.6 km/s and laps the Earth in 90 minutes. A far-out GEO satellite ambles at just 3.1 km/s and takes a full day. Put the low, middle, and high rings on screen together and the difference is impossible to miss: the inner ring blurs past while the outer ring barely seems to budge.

Predict first: Before you turn them on: three orbits at three different heights, all racing at once. Do you expect the higher ones to move faster or slower — and does it feel odd that farther away would mean slower?

Try it in the simulator:

Turn on **LEO, GPS, and GEO** together and watch them move (raise time warp with **.**). Try slowing time back down with **,** and speeding it up again — at every rate the ordering holds: inner fast, outer slow.

What to look for: the low LEO ring whips around fast while the far-out GEO ring crawls, with GPS in between.

While you watch, think about:

- Line them up: does each ring get slower as it gets higher, without exception?
- Does it surprise you that being farther away makes an orbit slower, not faster?
- Roughly how many LEO laps go by in the time GEO completes a single one?

Check your understanding: Compared to a low LEO satellite, a satellite far out at GEO moves...

- Faster
- Slower
- The same speed
- Backwards

PART F · Geosynchronous orbit

Exercise 19 — Geosynchronous & geostationary

You've seen that GEO takes exactly one day per lap. Now watch why that one fact is so useful. If a satellite circles the Earth once per day, and the Earth also turns once per day, the two move in perfect step — from the ground the satellite appears to **hang motionless** over a single spot. This “once per day” property is called **geosynchronous**.

There's a finer point in the names. **Geosynchronous** just means a 24-hour period. **Geostationary** means geosynchronous *and* sitting right over the equator — only then does it truly hover over one fixed point. A geosynchronous satellite parked over a high-latitude place like Colorado still keeps that spot's longitude, but it rides low in the southern sky rather than straight overhead, which is why ground antennas there point at a slant.

Predict first: Before you watch: a GEO satellite circles once per day and the Earth turns once per day. What do you think the satellite will look like from the ground as the planet spins beneath it?

Try it in the simulator:

Turn on the **GEO lock demo**. Raise time warp and watch the red satellite over Colorado as Earth spins. For contrast, briefly turn on the **LEO demo** and notice how that low, fast

satellite races across the ground below instead of holding station.

What to look for: the GEO satellite stays locked above the same longitude — it circles once per day, matching Earth's spin.

While you watch, think about:

- The satellite is racing along at ~ 3 km/s, so why does it look like it's standing still from the ground?
- What would break the illusion — a slightly faster orbit, or a slightly slower one?
- Why does a satellite over Colorado sit low in the southern sky instead of straight overhead?

Check your understanding: A geostationary satellite appears to hover over one spot on Earth because...

- It is not moving at all — it simply sits still above its chosen spot
- Its period equals one Earth day, so it keeps pace with the ground below
- It sits beyond the reach of Earth's gravity, so nothing pulls it off station
- It fires its thrusters continuously to hold itself above that one point

PART G · Inclination & launch geography

Exercise 20 — Inclination: tilting the orbit

Every orbit lies in a flat plane slicing through the center of the Earth. To say how an orbit is tilted, you need something fixed to measure the tilt *against* — and the reference everyone uses is **Earth's equatorial plane**: imagine the equator extended outward into space as a vast flat sheet, sitting exactly perpendicular to Earth's spin axis. **Inclination is the angle between the orbit's plane and that sheet.**

That one number tells you a lot. At 0° the orbit lies *in* the equatorial sheet — the satellite rides around directly above the equator forever. Tilt the plane and the satellite spends each lap swinging above the sheet and back below it — and here's the handy rule of thumb: **the inclination is also the highest latitude the satellite ever reaches.** A 30° orbit gets no farther north or south than 30° ; a 53° Starlink orbit tops out near Seattle's latitude; only a steep tilt can carry a satellite over your head if you live far from the equator.

The extreme case is a **90° (polar) orbit**, which passes right over the North and South Poles. As the satellite loops pole-to-pole, the Earth keeps spinning underneath it, so over time the satellite eventually **overflies every point on the planet**. That's exactly what you want for mapping, imaging, and weather satellites — and it's the same reason the Starlink and GPS shells you saw earlier were tilted, to reach beyond the equator.

Predict first: Before you move the slider: inclination tilts the orbit's plane. What do you think happens to how far north and south the satellite travels as you crank the tilt up toward 90° ?

Try it in the simulator:

Set the **inclination slider** to $\sim 0^\circ$ and inject to see an equatorial orbit, then raise it to $\sim 60^\circ$ and inject again, then to 90° (polar). Compare all three — watch the orbit reach higher and higher latitudes until, at 90° , it goes straight over the poles.

What to look for: the orbit plane tilts away from the equatorial plane as inclination rises — and the highest latitude each orbit reaches matches its inclination angle — until, at 90°, the orbit passes directly over the poles.

While you watch, think about:

- As you raise the slider, how far north and south does the orbit start to reach?
- What does a 90° polar orbit eventually let a satellite see that an equatorial one never will?
- How does this connect to why the Starlink and GPS shells were tilted?

Check your understanding: A satellite in a 90° (polar) orbit is useful because...

- It hovers over one chosen city, the way a geostationary satellite does
- It eventually passes over every latitude on Earth, the poles included
- It needs no propellant, because a polar plane never has to be corrected
- It is geostationary, since a 90° tilt still matches the Earth's rotation

Exercise 21 — Why launch sites differ (Vandenberg vs. Florida)

Where and which direction a rocket launches isn't arbitrary — it comes straight from the physics and safety you've been learning. Two forces set the rules. First, launching **east** banks Earth's spin for a free speed boost, which is why Florida sends most missions eastward out over the Atlantic for low-inclination orbits. Second, and non-negotiable: a rocket must not fly over populated land, in case something goes wrong.

That second rule is why **polar** missions launch from **Vandenberg** in California, heading **south**. A due-south track from Vandenberg runs down the open Pacific — empty ocean the whole way. The very same southward heading from Florida would cross populated land. Same desired orbit, different site, chosen so the ground track stays over water.

Predict first: Before reading on: picture a US map. If you had to launch a rocket south for a polar orbit and keep its flight path off populated land, would you pick a California coast site or a Florida one — and why?

Try it in the simulator:

Think about launch direction: trace a due-south path from Vandenberg (open Pacific) versus from Florida (populated land), and trace an eastward path from Florida (out over the Atlantic, banking Earth's spin). Compare what lies under each track.

What to look for: the safe launch heading is the one whose ground track stays over open ocean rather than crossing populated land.

While you watch, think about:

- For a polar launch, what lies south of Vandenberg versus south of Florida?
- Why does launching east, rather than west, give a rocket a running start?
- If you had to pick a launch site for a new polar-imaging satellite, what would you look for on the map?

Check your understanding: Why do polar-orbit missions launch from Vandenberg (CA) toward the south, not from Florida?

- California's weather gives more launch days per year than Florida's does

- A due-south track from Vandenberg flies over open ocean, not populated land
- Polar orbits must be entered westward, which Florida's geography forbids
- Vandenberg's higher elevation means less atmosphere to climb through

Key points

- An orbit is just **falling sideways fast enough to keep missing the Earth**. Too slow → you hit the ground; too fast → you escape.
- At a given altitude there is **one circular speed**. Release slower and that point becomes **apogee** (high point); faster and it becomes **perigee** (low point).
- On an ellipse an object moves **fastest at perigee (closest), slowest at apogee (farthest)**.
- **Farther out = slower**. LEO ~7.6 km/s (90 min); GEO ~3.1 km/s (24 h).
- **Prograde** = with Earth's spin (the cheap, default direction). **Retrograde** = against it.
- **Geosynchronous** = ~24 h period. **Geostationary** = geosynchronous *and* over the equator, so it hangs above one spot.
- **Inclination** is the tilt of the orbit plane measured against **Earth's equatorial plane**; it equals the highest latitude the satellite reaches, and a 90° (polar) orbit eventually overflies every latitude.
- Launch geography follows physics & safety: **east from Florida** banks Earth's spin; **south from Vandenberg** reaches polar orbits over open ocean.
- Scale reality check: the **ISS skims the surface** (~420 km, ~7% of an Earth radius) — “space” is only a NYC-hop straight up.

Final quiz

Answers are checked in the online version.

Q1. What actually keeps a satellite from falling to Earth?

- Gravity is negligible that far up, so there is nothing left to fall toward
- It is moving sideways fast enough to keep missing the ground as it falls
- Its engines run continuously, holding it up against the pull of gravity
- Earth's magnetic field grips the satellite and holds it at that altitude

Q2. You release an object at the exact circular speed for its altitude. If you instead released it a bit FASTER, what would happen?

- It would trace a wider ellipse, with the release point becoming perigee
- It would hold the same circle, since altitude alone decides the orbit
- It would crash almost at once, as extra speed drives it into the Earth
- Nothing would change until drag eventually slowed it back to circular

Q3. On a stretched-out elliptical orbit, where does the satellite move fastest?

- At apogee, the farthest point
- At perigee, the closest point
- It moves at the same speed the whole way around
- Exactly halfway between the two

Q4. Why does a geostationary satellite appear to hang over one spot on Earth?

- It has stopped moving entirely, hanging motionless above its own longitude
- Its orbit takes exactly one day, so it keeps pace with the ground below
- It has flown beyond Earth's gravity, so nothing pulls it out of position
- It orbits much lower than the ISS, where the ground appears to move less

Q5. Why do polar-orbit missions launch south from Vandenberg in California rather than from Florida?

- California's weather gives more usable launch days than Florida's does
- A due-south track from Vandenberg flies over open ocean, not populated land
- Polar orbits have to be entered toward the west, which Florida cannot do
- Vandenberg sits at a higher elevation, so there is less air to climb through

Learn more

- Orbital mechanics
https://en.wikipedia.org/wiki/Orbital_mechanics
- Orbit
<https://en.wikipedia.org/wiki/Orbit>
- Kepler's laws of planetary motion
https://en.wikipedia.org/wiki/Kepler%27s_laws_of_planetary_motion
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- History of astronomy / celestial mechanics
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- International Space Station
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- Global Positioning System (GPS)
https://en.wikipedia.org/wiki/Global_Positioning_System
- Geostationary orbit
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